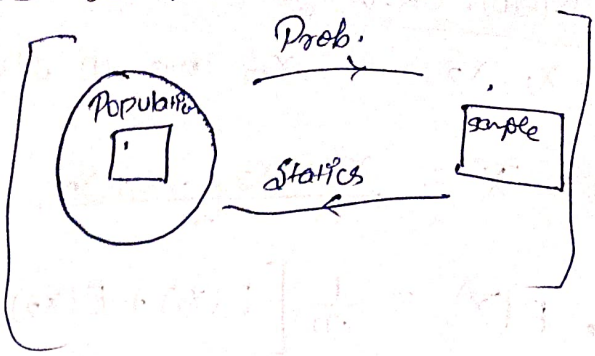
Statistics (Ch-24,25)

Advance Math Book



- Collection of data
- Analysis of data
- Arrangement, Selection, Conclude

Estimate of Parameter

↳ (Max. Likelihood Estimate) Method.

Confidence Interval

→ Testing of Hypothesis

→ (Regression Analysis)

Date :- 27/10/29

Estimation of Parameters :-

In probability theory, the mathematical model of all process are affected by chance, Whereas, in statistical theory these models are checked against observe real data.

Sample : The statistical is done by sampling i.e by drawing random sample from data set.

In this,

- Sample mean ($\bar{x}$)
- Sample Variance (S^2)

Note :- for any sample $x_1, x_2, \ldots, x_n$ of size 'n' the mean and the sample mean are same.

$$\mu = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n} = \bar{x}$$

(99) Variance $(\sigma^2) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$ $\quad [S^2 > \sigma^2]$

Sample variance $(S^2) = \frac{1}{n-1}\sum_{i=0}^{n}(x_i - \bar{x})^2$

Point Estimation :-

For any prob. func, there is one or more parameters are there, These parameters can be stimated by using the sample point $x_1, x_2, \ldots, x_n$ with the help of a method called Maximum Likelihood Estimate (MLE).

MLE :- (In General)

Let 'x' be any Random variable with prob. function is $f(x)$ depends on a single parameter θ.

i.e $f(x) = f(x, \theta)$ $\quad [x \rightarrow$ may be discrete or continuous$]$

Step 1 :- for n independent samples $x_1, x_2, \ldots, x_n$. we have $f(x_i) = f(x_i, \theta)$ for all $i = 1, 2, \ldots, n$.

Step 2 :- Define the likelihood function by

$$L = \prod_{i=1}^{n} f(x_i, \theta) = f(x_1, \theta) \cdot f(x_2, \theta) \ldots f(x_n, \theta)$$

Step 3 :- $\ln L = \ln f(x_1, \theta) + \ln f(x_2, \theta) + \ldots + \ln f(x_n, \theta)$

Step 4 :- for stationary point

$$\frac{\partial}{\partial \theta} \ln L = 0 \Rightarrow \theta = g(x_1, x_2, \ldots, x_n)$$

Generalization :-

Step 1, 2, 3 :- $f(x) = f(x, \underbrace{\theta_1, \theta_2, \ldots, \theta_n}_{\text{parameter}})$

Step 4 :- for extreme pt

$$\frac{\partial}{\partial \theta_1} \ln L = 0 \Rightarrow \theta_1 = g_1(x_1, \ldots, x_n)$$

$$\frac{\partial}{\partial \theta_2} \ln L = 0 \Rightarrow \theta_2 = g_2(x_1, \ldots, x_n)$$

$$\vdots$$

$$\frac{\partial}{\partial \theta_n} \ln L = 0 \Rightarrow \theta_n = g_n(x_1, \ldots, x_n).$$

Binomial, estimated 'p' not come n.

Normal either mean of normal population.

not come

$\frac{\partial}{\partial \sigma^2} \ln L = 0$ $\quad$ $\frac{\partial}{\partial \mu} \ln L = 0$

Q. Estimation the parameter of poisson distribution.

Date: 28/10/25

The PMF of Poisson distribution is

$$f(x,H) = \frac{e^{-H}.H^{x}}{x!} \quad ; x=0,1,2,3,\ldots$$

For n independent samples $x_1, x_2, \ldots, x_n$. we have

$$f(x_1,H) = \frac{e^{-H}.H^{x_1}}{x_1!}$$

$$f(x_2,H) = \frac{e^{-H}.H^{x_2}}{x_2!}$$

$$\vdots$$

$$f(x_n,H) = \frac{e^{-H}.H^{x_n}}{x_n!}$$

define the likelihood function by

$$L = \prod_{i=1}^{n} f(x_i,H)$$

$$= \frac{e^{-H}.H^{x_1}}{x_1!} \cdot \frac{e^{-H}.H^{x_2}}{x_2!} \cdot \cdots \cdot \frac{e^{-H}H^{x_n}}{x_n!}$$

$$= \frac{e^{-nH}\ H^{x_1+x_2+\cdots+x_n}}{x_1!.x_2!\ldots x_n!}$$

$$\ln L = -nH + \left(\sum_{i=1}^{n} x_i\right)\ln H - \ln(x_1!.x_2!\ldots x_n!)$$

For Extreme points

$$\frac{\partial}{\partial H}\ln L = 0$$

$$\Rightarrow -n + \left(\sum_{i=1}^{n} x_i\right)\frac{1}{H} = 0$$

$$\Rightarrow \boxed{\hat{H} = \frac{\sum_{i=1}^{n} x_i}{n} = \bar{x}}$$

6) Estimate the parameter of exponential distribution.

The PDF of exponential dist^n is

$$f(x,\lambda) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

For n independent samples $x_1, x_2, \ldots, x_n$, we have

$$f(x_1,\lambda) = \lambda e^{-\lambda x_1}$$
$$f(x_2,\lambda) = \lambda e^{-\lambda x_2}$$
$$\vdots$$
$$f(x_n,\lambda) = \lambda e^{-\lambda x_n}$$

Define the likelihood function by

$$L = \prod_{i=1}^{n} f(x_i,\lambda)$$
$$= \lambda^n \left[e^{-\lambda x_1} \cdot e^{-\lambda x_2} \cdot \ldots e^{-\lambda x_n} \right]$$
$$= \lambda^n e^{-\lambda \left(\sum_{i=1}^{n} x_i\right)}$$

$$\ln L = n \ln \lambda - \lambda \left(\sum_{i=1}^{n} x_i\right)$$

For extreme point.

$$\frac{\partial}{\partial \lambda} \ln L = 0$$

$$\Rightarrow \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0$$

$$\Rightarrow \frac{1}{\lambda} = \frac{\sum_{i=1}^{n} x_i}{n} \Rightarrow \boxed{\lambda = \frac{1}{\bar{x}}}$$

Q) Estimate the prob. of success for Binomial distribution on p.

$$\mu = \bar{x} = np \quad \Rightarrow p = \frac{\bar{x}}{n} \qquad (\text{respect on } p)$$

The PMF of Binomial distribution.

$$f(x; n, p) = \begin{cases} {}^{n}C_x\, p^x q^{n-x} & ; x = 0, \ldots, n \\ 0 & ; \text{otherwise} \end{cases}$$

For n independent sample

$$f(x_1; n, p) = {}^{n}C_{x_1}\, p^{x_1} q^{n-x_1}$$
$$f(x_2; n, p) = {}^{n}C_{x_2}\, p^{x_2} q^{n-x_2}$$
$$\vdots \qquad\qquad \vdots$$
$$f(x_n; n, p) = {}^{n}C_{x_n}\, p^{x_n} q^{n-x_n}$$

Define the likelihood function by

$$L = \prod_{i=1}^{n} f(x_i; n, p)$$
$$= \left[\left({}^{n}C_{x_1}\, p^{x_1} q^{n-x_1} \right) \cdot \left({}^{n}C_{x_2}\, p^{x_2} q^{n-x_2} \right) \cdot \ldots \left({}^{n}C_{x_n}\, p^{x_n} q^{n-x_n} \right) \right]$$
$$= \left({}^{n}C_{(x_1 + x_2 + \ldots x_3)}\, p^{(x_1 + x_2 + \ldots + x_n)}\, q^{(n-x_1)+(n-x_2)+\ldots+(n-x_n)} \right]$$

$$\ln(L) = \ln {}^{n}C_{\Sigma x_n} + \sum x_n \ln p + (n - \Sigma x_n) \ln (1-p)$$

$$\frac{\partial}{\partial p} \ln L = 0$$

$$\Rightarrow \Sigma x_n \ln p + (n - \Sigma x_n) \ln (1-p) = 0$$

$$\Rightarrow \frac{\Sigma x_n}{p} - \frac{n - \Sigma x_n}{1-p} = 0$$

$$\Rightarrow \frac{\Sigma x_n}{p} = \frac{n - \Sigma x_n}{1-p} \Rightarrow -\Sigma x_n (1-p) = np - p\Sigma x_n$$
$$\Rightarrow -\Sigma x_n + p\Sigma x_n = np - p\Sigma x_n$$
$$\Rightarrow \left[p = \frac{\Sigma x_i}{n} \right] \left[p = \bar{x} \right]$$

Estimate the parameter of normal distribution.

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad ; \quad -\infty < x < \infty$$

For n

$$f(x_1;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{-1}{2}\left(\frac{x_1-\mu}{\sigma}\right)^2}$$
$$f(x_2;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{-1}{2}\left(\frac{x_2-\mu}{\sigma}\right)^2}$$
$$\vdots$$
$$f(x_n;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{-1}{2}\left(\frac{x_n-\mu}{\sigma}\right)^2}$$

Define

$$L = \prod_{i=1}^{n} f(x_n;\mu,\sigma)$$
$$= \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\sigma}\right)^n \left(e^{\frac{-1}{2}\left\{\frac{\sum_{i=1}^{n} x_i - \mu}{\sigma}\right\}^2}\right)^n$$

$$\ln L = n\ln\left(\frac{1}{\sqrt{2\pi}}\right) + n\ln\left(\frac{1}{\sigma}\right) - \frac{n}{2}\left(\frac{\sum x_i - \mu}{\sigma}\right)^2$$

Estimation of Mean

$$\frac{\partial}{\partial \mu}\ln L = 0$$

Estimation of σ^2

$$\frac{\partial}{\partial \sigma^2}\ln L = 0$$

For extreme point,

$$\frac{\partial}{\partial \mu}\ln L = 0$$

$$\Rightarrow -\frac{n}{2}\left(\frac{\sum x_i^2 + \mu^2 - 2(\sum x_i)\mu}{\sigma^2}\right)$$

$$\Rightarrow -\frac{n}{2\sigma^2}(2\mu - 2\sum x_i) = 0$$

$$\Rightarrow \mu = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$(\mu = \bar{x})$$

$$\frac{\partial}{\partial \sigma^2}\ln L = 0$$

$$-\frac{n}{\sigma} + n\frac{(\sum x_i - \mu)^2}{\sigma} = 0$$

$$n\left(-1 + (\sum x_i - \mu)^2\right) = 0$$

$$(\sum x_i - \mu)^2 = 0$$

$$\ln L = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$-\frac{n}{2}\frac{1}{(\sigma^2)} + \frac{1}{2\sigma^4}\sum (x_i-\mu)^2$$

$$\sigma = -\frac{1}{2\sigma^2}(x_i-\mu)^2 = \frac{\partial}{\partial \mu}\ln L$$

$$\sum x_i = n\mu$$

$$\mu = \frac{\sum x_i}{n}$$

Regression Analysis :- (Not in Syllabus)

In statistical modelling, regression analysis is set of statistic progress for estimating the relationship b/w a dependent variable. (or outcome variable or response variable) to one or more independent variables (regressor or predictor variable).

The functional relationship of inde. variable with one or more ind. variable is called regression equation. The graph of regression eqn is called the regression curve. If the curve is a straight line then it called line of regression. (only this)

Note:-
If there are only two variables under consideration then is called simple regression.
If there more than two variables under consideration then the regression is called multiple regression.
If the relationship b/w x and y is non-linear then the regression is called Curvilinear regression.

Date:- 30/10/25

Regression Eqn of y on x :-

Let $(x_1, y_1), (x_2, y_2) \ldots (x_n, y_n)$ denote n pair of observation and suppose that corresponding straight line to be fitted to these data points. be

$$y = a + bx \quad -- (1)$$

Applying the least square method the normal eqn given by

$$na + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i \quad -- (2)$$

$$a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i \quad -- (3)$$

• Error = $y_{exact} - y_{app}$

• $q = \sum_{i=1}^{n} \left[y_i - (a + bx_i) \right]^2$

Proof (Not in Syllabus)

$\frac{\partial q}{\partial a} = 0$ | $\frac{\partial q}{\partial b} = 0$ is min

$\Rightarrow \Sigma (-2)(y_i - (a + bx_i)) = 0$

$\left[y_i = na + b\sum_{i=1}^{n} x_i \right]$

In eq (2)

$$a + b\frac{\Sigma x_i}{n} = \frac{\Sigma y_i}{n}$$

or, $a + b\bar{x} = \bar{y}$ ← (4)

Subtr. (3) & (4)

$$y - \bar{y} = b(x - \bar{x}) \quad -- (5)$$

$(\Sigma x_i) \times$ Eq^n (2) $- n \times$ eq^n (3)

$$b\left[\left(\sum_{i=1}^{n} x_i\right)^2 - n\sum_{i=1}^{n} x_i^2\right] = \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i - n\sum_{i=1}^{n} x_i y_i$$

or $$b = \frac{\Sigma x_i \Sigma y_i - n\Sigma x_i y_i}{(\Sigma x_i)^2 - n\Sigma x_i^2}$$

or $$b = \frac{n\Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

or $$b = \frac{\sum_{i=1}^{n} x_i y_i - \frac{(\Sigma x_i)(\Sigma y_i)}{n}}{\{\Sigma x_i^2 - \frac{(\Sigma x_i)^2}{n}}$$

Replace b by b_{yx}

$$b_{yx} = \frac{\sum_{i=1}^{n} x_i y_i - \frac{(\Sigma x_i)(\Sigma y_i)}{n}}{\{\sum x_i\}^2 - \frac{(\Sigma x_i)^2}{n}} \quad -- (6)$$

Eq^n 15) ⇒

$$y_i - \bar{y} = b_{yx}(x_i - \bar{x}) \quad -- 17)$$

Where b_{yx} is given eq^n. (6)

- Eq^n 7 is called the regression equation of y on x, and is used to estimate the value of y for a given value of x.

Regression equation of x on y :-

$$x = a' + b'y$$

[a' → Not derivative just for denotion]

Normal eqⁿ :-

$$na' + b' \Sigma y_i = \Sigma y_i$$

$$a' \Sigma y_i + b' \Sigma y_i^2 = \Sigma x_i y_i$$

The regression eqⁿ of x on y is

$$x_i - \bar{x} = b_{xy}(y_i - \bar{y})$$

[Derivation is not coming]

$$b_{xy} = \frac{\Sigma x_i y_i - \frac{(\Sigma x_i)(\Sigma y_i)}{n}}{\Sigma y_i^2 - \frac{(\Sigma y_i)^2}{n}}$$

Q) Find the regression eqⁿs for following data.

x	y	x^2	y^2	xy
5	20	25	400	100
10	30	100	900	300
15	40	225	1600	600
20	60	400	3600	1200
25	50	625	2500	1250
75	200	1375	9000	3450

$\bar{x} = 15$

$\bar{y} = 40$

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$$b_{yx} = \frac{\Sigma x_i y_i - \frac{(\Sigma x_i \Sigma y_i)}{n}}{\Sigma x_i^2 - \frac{(\Sigma x_i)^2}{n}} = \frac{3450 - \frac{15000}{5}}{1375 - \frac{5625}{5}} = \frac{\frac{9}{450}}{\frac{1250}{5}}$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$b_{xy} = \frac{450}{9000 - 800} = 0.05$$

$= \frac{9}{5}$ or 1.8

$$5a + 75b = 200$$

$$75a + 1375b = 3450$$

$$a = 1.8$$

$$b = 13$$

Ans

$$y - 200 = 1.8(x - 75)$$

$$y - 1.8x =$$

Regression eqn of x on y:

Reg. eq. is

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$\bar{x} = \frac{\Sigma x_i}{n} = \frac{75}{5} = 15$$

$$\bar{y} = \frac{\Sigma y_i}{n} = \frac{200}{5} = 40$$

$$b_{xy} = 0.45$$

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 15 = 0.45(y - 40)$$

$$x = 0.45y + 15 - 0.45 \times 40$$

$$= 15 + 0.45y - 18 = 0.45y - 3$$

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Q)

Use least square method to find the regression eqn of y on x for the following data.

x	-2	-1	0	1	2
y	1	2	3	3	4

x	y	x^2	y^2	xy
-2	1	4	1	-2
-1	2	1	4	-2
0	3	0	9	0
1	3	1	9	3
2	4	4	16	8
0	13	10	39	7

$$5a = 13 \Rightarrow \left[a = \frac{13}{5}\right]$$

$$10b = 7 \Rightarrow \left[b = \frac{7}{10}\right]$$

$$\frac{13}{5} + \frac{7}{10}\bar{x} = \bar{y}$$

$$2.6 + 0.7\bar{x} = \bar{y}$$

$$y = 2.6 + 0.7x$$

$$y = a + bx$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

Confidence Interval :-

↳ Confidence Interval for an unknown parameter θ of some dist^n are interval of the form $\theta_1 \le \theta \le \theta_2$ that contain θ not with certainity but with high prob. 'γ' i.e

$$CONF_\gamma \{ \theta_1 \le \theta \le \theta_2 \}$$

It means that $P(\theta_1 \le \theta \le \theta_2) = \gamma$

where θ_1 = lower confidence limit
θ_2 = upper confidence limit
γ = confidence level

(Confidence Interval for mean)

↳ with known variance → Using Normal dist^r. Table
↳ with unknown variance → t dist^r Table

Confidence Interval of μ (mean) of normal dist^r (known σ^2) :-

Algorithm

Step 1: Choose a confidence level γ (95%, 99%, 99.9%) [Not given then we take 95%]

Step 2: determine the value of c from formula

$$\phi(c) = \frac{1+\gamma}{2}$$ [using normal dist^n Table].

γ	0.90	0.95	0.99	0.999
c	1.645	1.96	2.576	3.291

Step 3: Calculate the mean ($\bar{x}$) from the given sample

$x_1, x_2, \dots, x_n$

Step 4: $CONF_\gamma \left(\bar{x} - \frac{c\sigma}{\sqrt{n}} \le \mu \le \bar{x} + \frac{c\sigma}{\sqrt{n}} \right)$

Date :- 03/11/25

5) Determine a 95% C.I. for μ of Normal population with variance 9, using a sample size 100 with sample mean 5.

Soln :- $\gamma = 0.95$

$$\phi(c) = \frac{1+\gamma}{2} \Rightarrow [c = 1.96] \leftarrow \text{In Table}$$

$$\phi(c) = 0.975$$

$\sigma^2 = 9 \Rightarrow [\sigma = 3]$, $\bar{x} = 5$

$$\text{CONF}_n \left(\bar{x} - \frac{c\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + \frac{c\sigma}{\sqrt{n}} \right)$$

$$\text{CONF}_n \left(5 - \frac{1.96 \times 3}{10} \leq \mu \leq 5 + \frac{1.96 \times 3}{10} \right)$$

$$\text{CONF}_n \left(4.412 \leq \mu \leq 5.588 \right)$$

6) Find 95% for mean of Normal population with Standard deviation 4 from the sample 39, 51, 49, 43, 57, 59

$\gamma = 0.95$

$\phi(c) = 0.975$

$c = 1.96$

$\bar{x} = 49.67$

$$\text{CONF}_n \left(49.67 - \frac{1.96 \times 4}{2.45} \leq \mu \leq 49.67 + \frac{1.96 \times 4}{2.45} \right)$$

$$\text{CONF}_n \left(46.846 \leq \mu \leq 52.87 \right)$$

Q) Find the sample size 'n' to obtain a C.I of length 0.3 of μ of normal population with $\sigma = 4$

Soln :- $CONF_\gamma\left(\bar{x} - \frac{C\sigma}{\sqrt{n}} \le \mu \le \bar{x} + \frac{C\sigma}{\sqrt{n}}\right)$

$$\text{Length} = 2\frac{C\sigma}{\sqrt{n}} \Rightarrow \sqrt{n} = 2\frac{C\sigma}{L} \Rightarrow n = \left(2\frac{C\sigma}{L}\right)^2$$

$$n = \left(\frac{2 \times 1.96 \times 4}{0.3}\right)^2$$

$$n = 2731.8$$

C.I for μ of Normal population (with σ^2 = unknown) :-

Step 1 :- Choose level γ

Step 2 :- Determine the value of C from formula

$$F(C) = \frac{1+\gamma}{2}$$ using (t-dist. Table)

with $(n-1)$ degree of freedom.

Step 3 :- Calculate $\bar{x}$ and s^2 from the given sample $\left[s^2 = \frac{1}{n-1}\Sigma(x_i - \bar{x})^2\right]$
(sample variance)

$x_1, x_2, \ldots, x_n$.

Step 4 :- $CONF_\gamma\left(\bar{x} - \frac{Cs}{\sqrt{n}} \le \mu \le \bar{x} + \frac{Cs}{\sqrt{n}}\right)$

Q) Assuming normality, determine a 99% C.I for the mean from the sample

144, 147, 146, 142, 144

~~Degree~~.

⇒ 4 check

$\gamma = 0.99 \Rightarrow F(C) = \frac{1+\gamma}{2} = 0.995$

$\bar{x} = 144.6$

$C = 4.6$

~~CONF$_\gamma$(144.6 ± 4.6x)~~

$$s^2 = \frac{1}{5-1}\left[(144-144.6) + (147-144.6) + (146-144.6) + (142-144.6) + (144-144.6)\right]$$

$s^2 = 3.8$

Date: 04/11/25

C.I for Variance of Normal Distribution :-

Algorithm

Step 1: Choose the γ^m

Step 2: Calculate the value of C_1 and C_2 by using the formula

$$F(C_1) = \frac{1-\gamma^m}{2} \quad \& \quad F(C_2) = \frac{1+\gamma^m}{2}$$ from

χ^2 – dist^r. table with $(n-1)$ degree of freedom.

↓ chi-square

Step 3: Calculate S^2 from the given sample $x_1, x_2, \ldots, x_n$

Step 4: $CONF_{\gamma^m}\left(\frac{(n-1)S^2}{C_2} \leq \sigma^2 \leq \frac{(n-1)S^2}{C_1} \right)$ ***

Q) Determine the 95% C.I for variance from the following sample

89, 84, 87, 81, 89, 86, 91, 90, 78, 89, 87, 99, 83, 89.

Soln:-

$$F(C_1) = \frac{1-0.95}{2} = 0.025 \qquad F(C_2) = \frac{1+0.95}{2} = 0.975.$$

$C_1 = 5.081$ $\qquad C_2 = 24.74$

$$S^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2 \qquad [\bar{x} = 87.3]$$

$$= \frac{1}{13}\Big[(89-87.3)^2 + (84-87.3)^2 + (87-87.3)^2 + (81-87.3)^2 + (89-87.3)^2 + (86-87.3)^2 + (91-87.3)^2 + (90-87.3)^2 + (78-87.3)^2 + (89-87.3)^2 + (87-87.3)^2 + (99-87.3)^2 + (83-87.3)^2 + (89-87.3)^2 \Big]$$

$$= \frac{1}{13}\Big[2.89 + 10.89 + 0.09 + 39.69 + 2.89 + 1.69 + 13.69 + 7.29 + 86.49 + 2.89 + 0.09 + 136.89 + 18.49 + 2.89 \Big]$$

$$\text{CONF}\left(\frac{323.86}{24.74} \leq \sigma^2 \leq \frac{323.86}{5.01} \right)$$

$$\text{CONF}\left(13.09 \leq \sigma^2 \leq \frac{(65225)}{64.64} \right)$$

NOTE :-

1) Let $X_1, X_2, \ldots, X_n$ be n ind. R.V with same mean (μ) and same variance (σ^2). Then R.V

$$\left[T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \right]$$ has t-distribution with $(n-1)$ degree of freedom.

(2) Let $X_1, X_2, \ldots, X_n$ be n inde. R.V with same mean (μ) and same variance (σ^2). Then R.V

$$\left[Y = \frac{(n-1)S^2}{\sigma^2} \right]$$ has χ^2 (chi-square) -distribution with $(n-1)$ d.o.f.

$$\left[Z_{\bar{x}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right]$$

↓ Standard Normal dist^n.

Ch → 25.4

Testing of Hypothesis :-

Significance Level :- (Error small)

↳ The significance level is defined by

$$[\alpha = 1 - \gamma]$$

↳ confidence Level.

↳ Testing of Hypothesis is a statistical test use to determine whether the hypothesis assumed is for the sample of data for true for the entire population.

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Ex → 25.3,

Mean (known value) or (unknown value)

Variance (known value) or unknown value

Date: 06/11/25

Testing of Hypothesis:-

Rejection Region :- (R.R)

Rejection Region or critical region is the region such that we reject the hypothesis if the observed value in the test in this region. So this is called Rejection Region.

Let θ be a unknown parameter in a dist^{r} and suppose that we want to test the (H) hypothesis $\theta = \theta_0$. ($\theta_0 \rightarrow$ fixed value) ($\theta \rightarrow$ parameter) then

there are three main times of alternative.

(i) $\theta > \theta_0$ — (1)
(ii) $\theta < \theta_0$ — (2)
(iii) $\theta \neq \theta_0$ — (3)

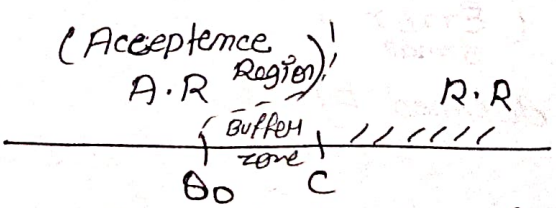
[Null Hyp. (H_0) we take for test of hypothesis]

Ex: $H_0 : \mu \leq 13$ $\quad$ $H_1 : \mu > 13$ (alternative Term)

$H_0 : \mu \geq 13$ $\quad$ $H_1 : \mu < 13$ (alternative)

$H_0 : \mu = 12$ $\quad$ $H_1 : \mu \neq 12$ (alternative term)

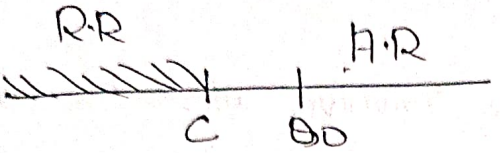
- one eq^{n} and two eq^{n} are called one-sided alternative and three are called two-sided alternative
- In one eq^{n} the critical C lies to the right of θ_0 so does the alternative. Hence the R.R extend to the right called right-sided test.



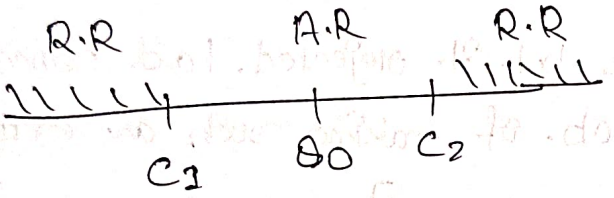
Eg → [$\mu \leq 300$, $\mu > 300$] Left cable

↳ act like a boundary behave b/w A.R and R.R.

- In two eq^{n} the critical C lies to the left of θ_0. so does the alternative. Hence the R.R extend to the left called left-sided test.



- In 3 eqn we have two R.R and hence it is called two-sided test.



Algorithm :-

Step 1 :- Formulate the Null hypothesis (H_0) to be tested (say)

$$H_0 : \theta = \theta_0$$

Step 2 :- Formulate the alternative i.e. (say)

$$H_1 : \theta_0 = \theta_1$$

Step 3 :- Choose the significance level (α) (usually 5%, 1% or 0.1%).

Step 4 :- Use a R.V $\hat{\theta} = g(x_1, x_2, \ldots, x_n)$ (In general 5%)

whose dist^n depends on a hypothesis and on alternative. and this dist^r is known bothe the cases.

Determine the critical value of C from the dist^n $\hat{\theta}$. Assuming the null hypothesis to be true.

Step 5 :- Use the sample $(x_1, x_2, \ldots, x_n)$ to the determine the observed value of $\hat{\theta}$.

Step 6 :- Accept all reject the hypothesis depending on size of $\hat{\theta}$ relative to C.

Errors in the Test :-

Testing of hypothesis always involve risk of making false decision.

Type-1 Error :- (α)

(Rejecting the true hypothesis)

↳ The hypothesis is true but it rejected. (and hence alternative is accepted). The prob. of making such an error equal to

$$\left[P(\hat{\theta} > C)_{\theta=\theta_0} = \alpha\right]$$

Date :- 07/10/25

Type-2 Error :- (β)

↳ The hypothesis is false but it accepted because $\hat{\theta}$ assumes a value $[\hat{\theta} \leq C]$ the prob. of making such an error is defined by $\left[P(\hat{\theta} \leq C)_{\theta=\theta_1} = \beta\right]$

Note :-

$\gamma = 1-\beta$ is called power of the test. Clearly, The powers of test is prob. of

Q) X be a R.V with variance 9 using a sample size of $n = 10$ with mean $\bar{X}$, test the hypothesis $H = 24$ against all type of hypothesis. of three of hypothesis

Soln :- $\sigma^2 = 9$, $n = 10$

$H_0 : \mu = 24 \quad (\theta = \theta_0)$

$\alpha = 0.05 \ (5\%)$

$$\bar{X} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$$

↓

μ and $\frac{\sigma^2}{n}$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

- Rejection region will be obtained from $\bar{x}$ if the hypothesis is true then $\bar{x}$ is normal with mean = 24 and variance

$$\frac{\sigma^2}{10} = 0.9$$

Case - I (Right sided test) $H_1 : \mu > 24 \quad (\theta > \theta_0)$

determine the c from prob. of c

$$P(\bar{x} > c)_{\mu=24} = \alpha$$

$$1 - P(\bar{x} \le c)_{\mu=24} = \alpha$$

$$P(\bar{x} \le c)_{\mu=24} = \alpha \; 1 - \alpha$$

$$P\left(z \le \frac{c-24}{\sqrt{0.9}}\right) = 1 - \alpha$$

$$\Rightarrow \Phi\left(\frac{c-24}{\sqrt{0.9}}\right) = 0.95 \longrightarrow \text{Normal dist}^{n} \text{ Table}$$

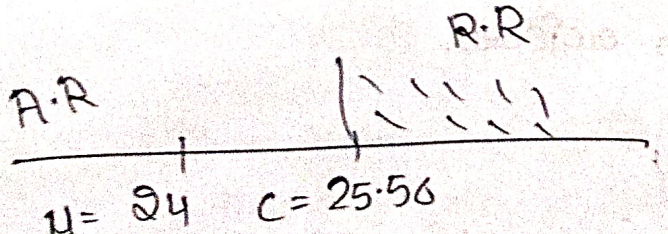
$$\frac{c-24}{\sqrt{0.9}} = 1.645$$

$$\Rightarrow c = 1.645(\sqrt{0.9}) + 24$$

$$\Rightarrow c = 1.56 + 24$$

$$c = 25.56$$

When variance not given then we use T-dist^{n}



- If $\bar{x} < 25.56$ then the null hypothesis is accepted. $(\theta = \theta_0)$
- If $\bar{x} > 25.56$ then the null hypothesis is rejected. alternative is accepted.

Case-2 (Left sided test) $(H_1 : H < 24)$

In this case the critical c is obtained by

$$P(\bar{x} \le c) = \alpha$$

$$P\left(z \le \frac{c-24}{\sqrt{0.9}}\right) = \alpha$$

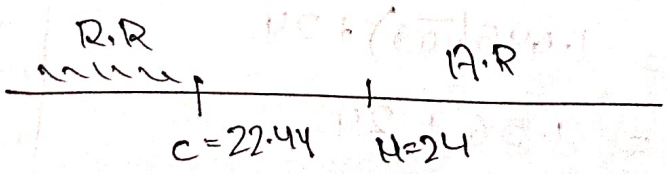
$$\phi\left(\frac{c-24}{\sqrt{0.9}}\right) = \alpha$$

$$\phi\left(\frac{c-24}{\sqrt{0.9}}\right) = 0.05$$

$$\frac{c-24}{\sqrt{0.9}} = -1.645$$

$$c = -1.56 + 24$$

$$c = 22.44$$



(Normal dist^r is symmetric)

If $\bar{x} > 22.44$ then H_0 is accepted

If $\bar{x} < 22.44$ then H_0 is rejected.

Case-3 :2 (Two-sided test)

Since, the normal dist^r is symmetric, we choice C_1 & C_2 equi-distance from H.

(Say) $C_1 = 24 - K$

$C_2 = 24 + K$

Now, we need to determine the value of k from

$$P(24-k \le \bar{x} \le 24+k) = 1-\alpha$$

~~$P\left(\frac{24-k-24}{\sqrt{0.9}} \le Z \le \frac{24+k-24}{\sqrt{0.9}}\right)$~~

$P(a \le X \le b)$

$$\Rightarrow \phi\left(\frac{b-\mu}{\sigma}\right) - \phi\left(\frac{a-\mu}{\sigma}\right)$$

$$\phi\left(\frac{24+k-24}{\sqrt{0.9}}\right) - \phi\left(\frac{24-k-24}{\sqrt{0.9}}\right) = 1-\alpha$$

$$\phi\left(\frac{k}{\sqrt{0.9}}\right) - \phi\left(\frac{-k}{\sqrt{0.9}}\right) = 1-\alpha$$

$$\phi\left(\frac{k}{\sqrt{0.9}}\right) - \left(1 - \phi\left(\frac{k}{\sqrt{0.9}}\right)\right) = 1-\alpha$$

$$2\phi\left(\frac{k}{\sqrt{0.9}}\right) - 1 = 0.95$$

$$2\phi\left(\frac{k}{\sqrt{0.9}}\right) = 1.95$$

$$\Rightarrow \phi\left(\frac{k}{\sqrt{0.9}}\right) = \frac{1.95}{2}$$

$$\Rightarrow \phi\left(\frac{k}{\sqrt{0.9}}\right) = 0.975$$

$$\frac{k}{\sqrt{0.9}} = 1.96$$

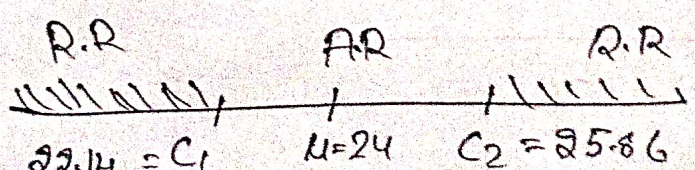
$$k = 1.96(\sqrt{0.9})$$

$$[k = 1.86]$$

$$C_1 = 24 - 1.86 = 22.14$$

$$C_2 = 24 + 1.86 = 25.86$$

$\bar{X} < 22.14$ & $\bar{X} > 25.86$ is accepted.



Date 10/11/25

Q) Suppose that you want to buy 100 coils of certain kind of wire. prove that you can verify the manufacturer claim

$\mu = 200lb$ and more. Suppose 25 coils are randomly selected and the breaking strength mean $\mu = 197$ and $\sigma = 6\ lb$. Test the hypothesis at the 5% level of significance.

In this θ. variance is not given

Left side test: $(\mu < 200\ lb) \Rightarrow H_0$

$H_1 : \mu < 200\ lb$

$P(\bar{X} \le c) = \alpha$

$T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}}$

$P\left(T \le \dfrac{c - 24}{6/5}\right) = 1 - \alpha$

$= 0.95$

Good Try

$H_0 \Rightarrow \mu = 200 \qquad \alpha = 0.05$

$H_1 \rightarrow \mu < 200$

$T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} = -2.5$ (✓) $\left[\dfrac{197 - 200}{6/\sqrt{25}} = -2.5\right]$

$P(\bar{X} \le c) = \alpha$

$F(c) = 0.05$ (Symmetric)

$\Phi(c) = -1.71$ (✓)

R.R | A.R

$c = -1.71$


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